

Modeling Song Popularity: Neural vs Bayesian

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Background, Motivation, and Goals

- Can audio features predict song popularity?
- Understanding popularity helps music platforms and artists
- Compare Neural Network and Bayesian models

Data Description

Dataset: Spotify Tracks Dataset (~114,000 songs)

Features: danceability, energy, tempo, loudness, valence

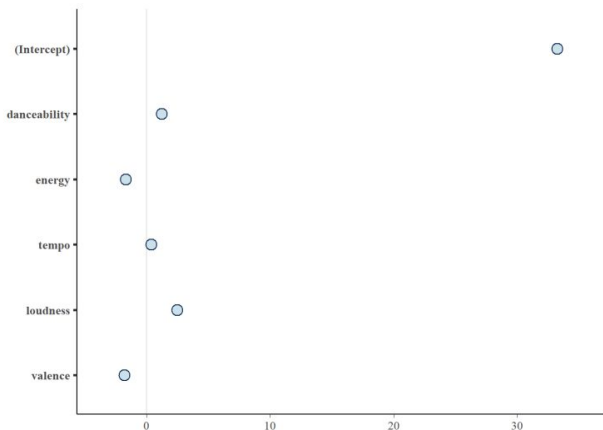
Target: Popularity (0-100)

Bayesian Regression (R)

Linear model with priors

Posterior estimation

Provides interpretability



Loudness has the strongest positive effect, while energy and valence are negative.

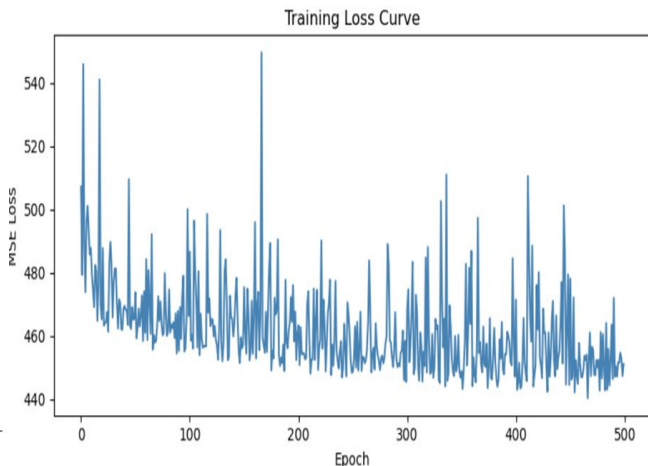
Neural Network (NumPy) Architecture: 5 → 64 → 32 → 1.

ReLU + Linear output.

Mini-batch SGD

Conclusion

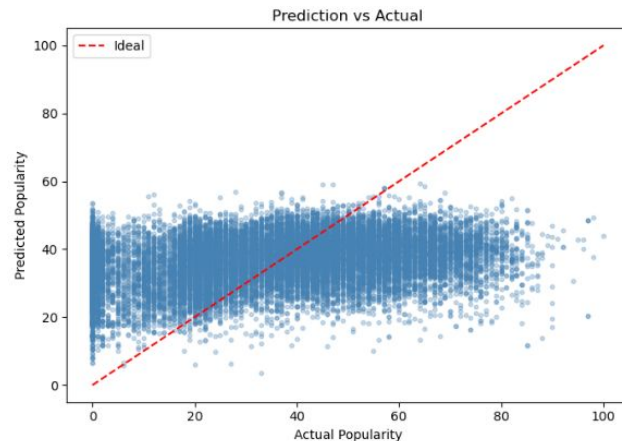
Both models suggest that audio features have limited predictive power, but provide useful insights into song characteristics. Future work could include incorporating external factors such as artist popularity, social media engagement, or playlist exposure.



Loss decreases over time, indicating the model is learning.

Findings:

- Predictions are concentrated around the mean, indicating regression toward the average
- The neural network shows limited predictive performance (low R^2)
- The Bayesian model provides better interpretability of feature effects
- Loudness and danceability have positive effects on popularity
- Audio features alone are not sufficient to fully explain song popularity



Predictions are concentrated around the mean, showing limited predictive power.